

Unit 5 UI Flutter:

Using Common Widgets

Container:

```
import 'package:flutter/material.dart';

void main() => runApp(MyApp());

/// This Widget is the main application widget.
class MyApp extends StatelessWidget {

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyContainerWidget(),
    );
  }
}

class MyContainerWidget extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text("Flutter Container Example"),
        ),
        body: Container(
          width: double.infinity,
          height: 150.0,
          color: Colors.green,
          margin: EdgeInsets.all(25),
          padding: EdgeInsets.all(35),
          alignment: Alignment.center,
          transform: Matrix4.rotationZ(0.1),
          child: Text("Hello! I am Sudhakar Reddy!!",
            style: TextStyle(fontSize: 25)),
        ),
      ),
    );
  }
}
```

Row Widget:

```
import 'package:flutter/material.dart';
```

```

void main() { runApp(MyApp()); }
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage()
    );
  }
}

```

```

class MyHomePage extends StatefulWidget {
  @override
  _MyHomePageState createState() => _MyHomePageState();
}

```

```

class _MyHomePageState extends State<MyHomePage> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Flutter Row Example"),
      ),
      body: Row(
        mainAxisAlignment: MainAxisAlignment.spaceEvenly,
        children:<Widget>[
          Container(
            margin: EdgeInsets.all(12.0),
            padding: EdgeInsets.all(8.0),
            decoration:BoxDecoration(
              borderRadius:BorderRadius.circular(8),
              color:Colors.green
            ),
            child: Text("React.js",style: TextStyle(color:Colors.yellowAccent,fontSize:25),),
          ),
          Container(
            margin: EdgeInsets.all(15.0),
            padding: EdgeInsets.all(8.0),
            decoration:BoxDecoration(
              borderRadius:BorderRadius.circular(8),
              color:Colors.green
            ),
            child: Text("Flutter",style: TextStyle(color:Colors.yellowAccent,fontSize:25),),
          ),
          Container(
            margin: EdgeInsets.all(12.0),
            padding: EdgeInsets.all(8.0),
            decoration:BoxDecoration(
              borderRadius:BorderRadius.circular(8),
              color:Colors.green
            ),

```

```

        child: Text("MySQL",style: TextStyle(color:Colors.yellowAccent,fontSize:25),),
      ),
    ],
  ),
);
}
}

```

Column Widget:

```
import 'package:flutter/material.dart';
```

```
void main() { runApp(MyApp()); }
```

```
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyHomePage()
    );
  }
}

```

```
class MyHomePage extends StatefulWidget {
  @override
  _MyHomePageState createState() => _MyHomePageState();
}

```

```
class _MyHomePageState extends State<MyHomePage> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Flutter Column Example"),
      ),
      body: Column(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children:<Widget>[
          Container(
            margin: EdgeInsets.all(20.0),
            padding: EdgeInsets.all(12.0),
            decoration:BoxDecoration(
              borderRadius:BorderRadius.circular(8),
              color:Colors.red
            ),
            child: Text("React.js",style: TextStyle(color:Colors.yellowAccent,fontSize:20),),
          ),
          Container(
            margin: EdgeInsets.all(20.0),
            padding: EdgeInsets.all(12.0),

```

```

        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(8),
          color: Colors.red
        ),
        child: Text("Flutter", style: TextStyle(color: Colors.yellowAccent, fontSize: 20)),
      ),
      Container(
        margin: EdgeInsets.all(20.0),
        padding: EdgeInsets.all(12.0),
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(8),
          color: Colors.red
        ),
        child: Text("MySQL", style: TextStyle(color: Colors.yellowAccent, fontSize: 20)),
      )
    ],
  ),
);
}
}

```

Text Widget:

```
import 'package:flutter/material.dart';
```

```
void main() { runApp(MyApp()); }
```

```

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        primarySwatch: Colors.green,
      ),
      home: MyTextPage()
    );
  }
}

class MyTextPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Text Widget Example")
      ),
      body: Center(
        child: Text(
          "Hello World! This is a Text Widget.",
          style: TextStyle(

```

```

        fontSize: 35,
        color: Colors.purple,
        fontWeight: FontWeight.w700,
        fontStyle: FontStyle.italic,
        letterSpacing: 8,
        wordSpacing: 20,
        backgroundColor: Colors.yellow,
        shadows: [
          Shadow(color: Colors.blueAccent, offset: Offset(2,1), blurRadius:10)
        ],
      ),
    ),
  ),
);
}
}

```

Rich Text Widget:

```

import 'package:flutter/material.dart';
import 'package:flutter/gestures.dart';

void main() { runApp(MyApp()); }

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      theme: ThemeData(
        primarySwatch: Colors.green,
      ),
      home: MyTextPage()
    );
  }
}

class MyTextPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Rich Text Widget Example")
      ),
      body: Container(
        child: Center(
          child: RichText(
            text: TextSpan(
              text: 'Don\'t have an account?',
              style: TextStyle(color: Colors.black, fontSize: 20),
              children: <TextSpan>[

```

```

        TextSpan(text: ' Sign up',
          style: TextStyle(color: Colors.blueAccent, fontSize: 20),
          recognizer: TapGestureRecognizer()
            ..onTap = () {}
        )
      ],
    ),
  ),
)
);
}
}

```

Icon:

```

import 'package:flutter/material.dart';

void main() { runApp(MyApp()); }

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyTextPage()
    );
  }
}

class MyTextPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("Rich Text Widget Example")
      ),
      body: Container(
        child: Center(
          child: RichText(
            text: TextSpan(
              style: Theme.of(context).textTheme.body1,
              children: [
                TextSpan(text: 'Click ', style: TextStyle(fontSize: 25)),
                WidgetSpan(
                  child: Padding(
                    padding: const EdgeInsets.symmetric(horizontal: 2.0),
                    child: Icon(Icons.add, color: Colors.red),
                  ),
                ),
                TextSpan(text: ' to add', style: TextStyle(fontSize: 25)),
              ],
            ),
          ),
        ),
      ),
    );
  }
}

```

```

    ),
  ),
),
);
}
}

```

Text Field:

```

import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp( home: MyApp(),));
}

class MyApp extends StatefulWidget {
  @override
  _State createState() => _State();
}

class _State extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter TextField Example'),
      ),
      body: Padding(
        padding: EdgeInsets.all(15),
        child: Column(
          children: <Widget>[
            Padding(
              padding: EdgeInsets.all(15),
              child: TextField(
                decoration: InputDecoration(
                  border: OutlineInputBorder(),
                  labelText: 'User Name',
                  hintText: 'Enter Your Name',
                ),
              ),
            ),
            Padding(
              padding: EdgeInsets.all(15),
              child: TextField(
                obscureText: true,
                decoration: InputDecoration(
                  border: OutlineInputBorder(),
                  labelText: 'Password',
                  hintText: 'Enter Password',
                ),
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```

```

        ),
      ),
      RaisedButton(
        textColor: Colors.white,
        color: Colors.blue,
        child: Text('Sign In'),
        onPressed: () {},
      )
    ],
  ),
);
}
}

```

Flat Button:

```

import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Flutter FlatButton Example'),
        ),
        body: Center(child: Column(children: <Widget>[
          Container(
            margin: EdgeInsets.all(25),
            child: FlatButton(
              child: Text('SignUp', style: TextStyle(fontSize: 20.0),),
              onPressed: () {},
            ),
          ),
          Container(
            margin: EdgeInsets.all(25),
            child: FlatButton(
              child: Text('LogIn', style: TextStyle(fontSize: 20.0),),
              color: Colors.blueAccent,
              textColor: Colors.white,
              onPressed: () {},
            ),
          ),
        ]),
      ),
    );
  }
}

```



```

    ),
  ),
]
))
),
);
}
}

```

Raised Button:

```

import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  String msg = 'Flutter RaisedButton Example';
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Flutter RaisedButton Example'),
        ),
        body: Container(
          child: Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                Text(msg, style: TextStyle(fontSize: 30, fontStyle: FontStyle.italic),),
                RaisedButton(
                  child: Text("Click Here", style: TextStyle(fontSize: 20),),
                  onPressed: _changeText,
                  color: Colors.red,
                  textColor: Colors.yellow,
                  padding: EdgeInsets.all(8.0),
                  splashColor: Colors.grey,
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}

```

```

}
_changeText() {
  _setState() {
    if (msg.startsWith('F')) {
      msg = 'We have learned FlutterRaised button example.';
    } else {
      msg = 'Flutter RaisedButton Example';
    }
  }
});
}
}

```

Floating Action Button

```

import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(home: Scaffold(
      appBar: AppBar(
        title: Text("FAB Button Example"),
        backgroundColor: Colors.blue,
        actions: <Widget>[
          IconButton(icon: Icon(Icons.camera_alt), onPressed: () => {}),
          IconButton(icon: Icon(Icons.account_circle), onPressed: () => {}),
        ],
      ),
      floatingActionButton: FloatingActionButton(
        child: Icon(Icons.navigation),
        backgroundColor: Colors.green,
        foregroundColor: Colors.white,
        onPressed: () => {},
      ),
      /*floatingActionButton: FloatingActionButton.extended(
        onPressed: () {},
        icon: Icon(Icons.save),
        label: Text("Save"),
      ), */
    ));
  }
}

```

```
}  
}
```

Icon Button

```
import 'package:flutter/material.dart';  
  
void main() => runApp(MyApp());  
  
class MyApp extends StatelessWidget {  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: Scaffold(  
        appBar: AppBar(  
          title: Text("Icon Button Example"),  
        ),  
        body: Center(  
          child: MyStatefulWidget(),  
        ),  
      ),  
    );  
  }  
}  
  
double _speakervolume = 0.0;  
  
class MyStatefulWidget extends StatefulWidget {  
  MyStatefulWidget({Key key}) : super(key: key);  
  
  @override  
  _MyStatefulWidgetState createState() => _MyStatefulWidgetState();  
}  
  
class _MyStatefulWidgetState extends State<MyStatefulWidget> {  
  Widget build(BuildContext context) {  
    return Column(  
      mainAxisAlignment: MainAxisAlignment.min,  
      children: <Widget>[  
        IconButton(  
          icon: Icon(Icons.volume_up),  
          iconSize: 50,  
          color: Colors.brown,  
          tooltip: 'Increase volume by 5',  
          onPressed: () {  
            setState() {  
              _speakervolume += 5;  
            }  
          }  
        ),  
        Text('Speaker Volume: $_speakervolume')  
      ],  
    );  
  }  
}
```

```

    ],
  );
}
}

```

PopupMenu Button

```

import 'package:flutter/material.dart';

void main() { runApp(MyApp());}

class MyApp extends StatefulWidget {
  @override
  _MyAppState createState() => _MyAppState();
}

class _MyAppState extends State<MyApp> {
  Choice _selectedOption = choices[0];

  void _select(Choice choice) {
    setState(() {
      _selectedOption = choice;
    });
  }

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: const Text('PopupMenu Button Example'),
          actions: <Widget>[
            PopupMenuButton<Choice>(
              onSelect: _select,
              itemBuilder: (BuildContext context) {
                return choices.skip(0).map((Choice choice) {
                  return PopupMenuItem<Choice>(
                    value: choice,
                    child: Text(choice.name),
                  );
                }).toList();
              },
            ),
          ],
        ),
        body: Padding(
          padding: const EdgeInsets.all(10.0),
          child: ChoiceCard(choice: _selectedOption),
        ),
      ),
    );
  }
}

```

```

    }
  }

class Choice {
  const Choice({this.name, this.icon});
  final String name;
  final IconData icon;
}

const List<Choice> choices = const <Choice>[
  const Choice(name: 'Wi-Fi', icon: Icons.wifi),
  const Choice(name: 'Bluetooth', icon: Icons.bluetooth),
  const Choice(name: 'Battery', icon: Icons.battery_alert),
  const Choice(name: 'Storage', icon: Icons.storage),
];

class ChoiceCard extends StatelessWidget {
  const ChoiceCard({Key key, this.choice}) : super(key: key);

  final Choice choice;

  @override
  Widget build(BuildContext context) {
    final TextStyle textStyle = Theme.of(context).textTheme.headline;
    return Card(
      color: Colors.greenAccent,
      child: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.min,
          crossAxisAlignment: CrossAxisAlignment.center,
          children: <Widget>[
            Icon(choice.icon, size: 115.0, color: textStyle.color),
            Text(choice.name, style: textStyle),
          ],
        ),
      ),
    );
  }
}

```

Button Bar

```

import 'package:flutter/material.dart';

void main() {
  runApp(MaterialApp( home: MyApp(),));
}

class MyApp extends StatefulWidget {
  @override

```

```

    _State createState() => _State();
  }

class _State extends State<MyApp> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter ButtonBar Example'),
      ),
      body: Padding(
        padding: EdgeInsets.all(10),
        child: Column(
          children: <Widget>[
            Padding(
              padding: EdgeInsets.all(15),
              child: new ButtonBar(
                mainAxisAlignment: MainAxisAlignment.min,
                children: <Widget>[
                  RaisedButton(
                    child: new Text('Sudhakar Reddy'),
                    color: Colors.lightGreen,
                    onPressed: () {/** */},
                  ),
                  FlatButton(
                    child: Text('Flutter'),
                    color: Colors.lightGreen,
                    onPressed: () {/** */},
                  ),
                  FlatButton(
                    child: Text('MySQL'),
                    color: Colors.lightGreen,
                    onPressed: () {/** */},
                  ),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}

```

Displaying Images:

```
import 'package:flutter/material.dart';
```

```
void main() => runApp(MyApp());
```

```
class MyApp extends StatelessWidget {
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return MaterialApp(
```

```
      home: Scaffold(
```

```
        appBar: AppBar(
```

```
          title: Text('Flutter Image Demo'),
```

```
        ),
```

```
        body: Center(
```

```
          child: Column(
```

```
            children: <Widget>[
```

```
              Image.asset('assets/tablet.png'),
```

```
              Text(
```

```
                'A tablet is a wireless touch screen computer that is smaller than a notebook  
but larger than a smartphone.',
```

```
                style: TextStyle(fontSize: 20.0),
```

```
            ),
```

```
          ],
```

```
        ),
```

```
      ),
```

```
    );
```

```
  }
```

```
}
```

Displaying Icons:

```
import 'package:flutter/material.dart';
```

```
void main() => runApp(MyApp());
```

```
class MyApp extends StatelessWidget {
```

```
  // This widget is the root of your application.
```

```
  @override
```

```
  Widget build(BuildContext context) {
```

```
    return MaterialApp(
```

```
      theme: ThemeData(
```

```
        primarySwatch: Colors.blue,
```

```
      ),
```

```
      home: MyIconPage(),
```

```
    );
```

```
  }
```

```
}
```

```
class MyIconPage extends StatefulWidget {
```

```
  @override
```

```
  _MyIconPageState createState() => _MyIconPageState();
```

```
}
```

```

class _MylconPageState extends State<MylconPage> {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Flutter Icon Tutorial'),
      ),
      body: Row(
        mainAxisAlignment: MainAxisAlignment.spaceAround,
        children: <Widget>[
          Icon(Icons.camera_enhance),
          Icon(Icons.camera_front),
          Icon(Icons.camera_rear),
        ],
      );
    }
  }
}

```

Animation:

```

import 'package:flutter/animation.dart';
import 'package:flutter/material.dart';

```

```

void main() => runApp(MyApp());

```

```

class MyApp extends StatelessWidget {
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Animation',
      theme: ThemeData(
        // This is the theme of your application.
        primarySwatch: Colors.blue,
      ),
      home: MyHomePage(),
    );
  }
}

class MyHomePage extends StatefulWidget {
  _HomePageState createState() => _HomePageState();
}

class _HomePageState extends State<MyHomePage> with SingleTickerProviderState
Mixin {
  Animation<double> animation;
  AnimationController animationController;
  @override
  void initState() {
    super.initState();
    animationController = AnimationController(vsync: this, duration: Duration(milliseconds: 2500));
  }
}

```



```

animation = Tween<double>(begin: 0.0, end: 1.0).animate(animationController);
animation.addListener((){
  setState(){
    print (animation.value.toString());
  });
});
animation.addStatusListener((status){
  if(status == AnimationStatus.completed){
    animationController.reverse();
  } else if(status == AnimationStatus.dismissed) {
    animationController.forward();
  }
});
animationController.forward();
}
@override
Widget build(BuildContext context) {
  return Center(
    child: AnimatedLogo(
      animation: animation,
    )
  );
}
}
class AnimatedLogo extends AnimatedWidget {
  final Tween<double> _sizeAnimation = Tween<double> (begin: 0.0, end: 500.0);
  AnimatedLogo({Key key, Animation animation}):super(key: key, listenable: animation);

  @override
  Widget build(BuildContext context) {
    final Animation<double> animation = listenable;
    return Transform.scale(
      scale: _sizeAnimation.evaluate(animation),
      child: FlutterLogo(),
    );
  }
}

```